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|----------------------------------------------------------------------|---------------------------------------|
| Program : <b>Diploma in Cyber Forensics And Information Security</b> |                                       |
| Course Code : <b>5282</b>                                            | Course Title: <b>Network Security</b> |
| Semester: <b>5</b>                                                   | Credits: <b>4</b>                     |
| Course Category: Program Core Course                                 |                                       |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>                             | Periods per semester: <b>60</b>       |

### Course Objectives:

- To know the concepts & basic vocabulary of network security & security infrastructures.
- To identify the various cryptographic protocols along with hardware & software security
- To understand WAP security & security issues with WTLS.
- To learn how Network Access Controls can be implemented.
- To identify the application and use of Firewalls.

### Course Prerequisites:

| Topics                               | Course Code | Course name                                            | Semester |
|--------------------------------------|-------------|--------------------------------------------------------|----------|
| Basic knowledge in Operating Systems |             | Operating System                                       | III      |
| Basic knowledge in Cryptography      |             | Fundamentals of Cryptography and Information Security. | III      |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                                          | Duration (Hours) | Cognitive Level |
|-----------------|----------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Explain the features of Computer Networks, protocols, reference models and fundamental concepts of Network security. | 15               | Understanding   |
| CO2             | Understand the security concerns in Data link layer and the protocols for Wireless security and IP security.         | 14               | Understanding   |

|     |                                                                                                               |    |               |
|-----|---------------------------------------------------------------------------------------------------------------|----|---------------|
| CO3 | Describe the functioning of most common security protocols employed at Transport Layer and Application layer. | 15 | Understanding |
| CO4 | Understand the importance of Network Access Control and the role of Firewalls in Network security.            | 14 | Understanding |
|     | Series Test                                                                                                   | 2  |               |

#### CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   | 2   |     |     |     |     |     |
| CO3             | 3   | 2   |     |     |     |     |     |
| CO4             | 3   | 2   |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

#### Course Outline:

| Module Outcomes | Description                                                                                                                 | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Explain the features of Computer Networks, protocols, reference models and fundamental concepts of Network security.</b> |                  |                 |
| M1.01           | Explain computer networks classification based on the topology.                                                             | 2                | Understanding   |
| M1.02           | Explain computer networks classification based on the scale and the distance they cover.                                    | 2                | Understanding   |
| M1.03           | Explain Network Reference model, ISO/OSI reference model and TCP/IP reference model.                                        | 3                | Understanding   |
| M1.04           | Explain the OSI security architecture in detail.                                                                            | 2                | Understanding   |
| M1.05           | Explain the needs for Network security and the fundamental concepts of Network Security.                                    | 2                | Understanding   |
| M1.06           | Explain the goals of Network security. Describe the different types of Network security.                                    | 2                | Understanding   |
| M1.07           | Explain the basic model for Network Security.                                                                               | 2                | Understanding   |

Network Fundamentals : Computer Networks – Bus topology, Star topology, Ring topology , Tree topology, Mesh topology , Categories of Networks – LAN,MAN,WAN. Network Reference models – ISO/OSI reference model, TCP/IP reference model, The OSI security architecture.

Network Security Overview: Fundamental concepts, Goals of Network Security, Types of Network security, a model for Network Security.

|            |                                                                                                                           |   |               |
|------------|---------------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO2</b> | <b>Understand the security concerns in Data link layer and the protocols for Wireless security and IP security.</b>       |   |               |
| M2.01      | Explain the functionalities of Data Link Layer in the internet and the various security concerns in Data Link Layer.      | 2 | Understanding |
| M2.02      | Explain the different ways for Securing Ethernet LANs and Virtual LAN.                                                    | 2 | Understanding |
| M2.03      | Explain various wireless network threats and wireless security measures.                                                  | 2 | Understanding |
| M2.04      | Explain Wireless Access Protocol (WAP).                                                                                   | 1 | Understanding |
| M2.05      | Illustrate how to secure a Wireless LAN. Explain about Mobile device security and security concerns in the Network Layer. | 3 | Understanding |
| M2.06      | Explain IPsec, its Applications, Benefits, Services and different IPsec Communication Modes.                              | 2 | Understanding |
| M2.07      | Explain IPsec Protocols and IP security policy.                                                                           | 2 | Understanding |
|            | Series Test -1                                                                                                            | 1 |               |

Data Link Layer security: Functionalities of Data Link layer in the network, Security Concerns in Data Link Layer, Securing Ethernet LANs, Securing Virtual LAN, Wireless network threats, wireless security measures, WAP, Securing Wireless LAN, Mobile device security .

Network Layer security: Security in Network Layer, Overview of IPsec, Applications, Benefits, Services, IPsec Communication Modes, IPsec Protocols, IP security policy.

|            |                                                                                                                                          |   |               |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Describe the functioning of most common security protocols employed at Transport Layer and Application layer.</b>                     |   |               |
| M3.01      | Explain the Need for Transport Layer Security.                                                                                           | 1 | Understanding |
| M3.02      | Illustrate the role of Secure Socket Layer (SSL) and Transport Layer Security(TLS). Explain SSL and TLS architecture and protocol stack. | 2 | Understanding |
| M3.03      | Differentiate SSL and TLS. Discuss the various SSL/TLS attacks .                                                                         | 2 | Understanding |

|                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                       |   |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---|---------------|
| M3.04                                                                                                                                                                                                                                                                                                                                                    | Explain HTTPS-Connection initiation, Connection closure and Secure Shell Protocol (SSH).                              | 2 | Understanding |
| M3.05                                                                                                                                                                                                                                                                                                                                                    | Explain the security issues in the application layer,                                                                 | 1 | Understanding |
| M3.06                                                                                                                                                                                                                                                                                                                                                    | Explain the different threats that may affect Email and the measures to ensure E mail security.                       | 2 | Understanding |
| M3.07                                                                                                                                                                                                                                                                                                                                                    | Explain the factors to be considered for creating an e-mail policy and the steps involved.                            | 2 | Understanding |
| M3.08                                                                                                                                                                                                                                                                                                                                                    | Explain PGP and S/MIME in detail.                                                                                     | 2 | Understanding |
| M3.09                                                                                                                                                                                                                                                                                                                                                    | Illustrate how to ensure DNS security.                                                                                | 1 | Understanding |
| <p>Transport Layer security: Need for Transport Layer Security, Secure Socket Layer (SSL), TLS architecture, TLS Protocol, SSL/TLS attacks, HTTPS-Connection initiation, Connection closure, Secure Shell Protocol (SSH) .</p> <p>Application Layer security : E-mail Security – Email threats, Creating e-mail policy, PGP, S / MIME, DNS Security.</p> |                                                                                                                       |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                               | <b>Understand the importance of Network Access Control and the role of Firewalls in Network security.</b>             |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                    | Explain Network Access Control, the elements of Network Access Control system and Network access enforcement methods. | 3 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                    | Illustrate how Extensible Authentication Protocol work.                                                               | 1 | Understanding |
| M4.03                                                                                                                                                                                                                                                                                                                                                    | Explain IEEE 802.X Port based Network Access Control and how Access Control Lists for securing networks can be used.  | 3 | Understanding |
| M4.04                                                                                                                                                                                                                                                                                                                                                    | Explain Firewalls, the different types of Firewall and Stateless and Stateful Packet Filtering Firewall.              | 3 | Understanding |
| M4.05                                                                                                                                                                                                                                                                                                                                                    | Distinguish between Application Gateways and Circuit-Level Gateways.                                                  | 1 | Understanding |
| M4.06                                                                                                                                                                                                                                                                                                                                                    | Illustrate the Firewall Deployment with DMZ.                                                                          | 1 | Understanding |
| M4.07                                                                                                                                                                                                                                                                                                                                                    | Explain the security issues in internet banking system.                                                               | 2 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                          | Series Test – II                                                                                                      | 1 |               |

Network Access Control: Elements of a Network Access Control system , Network access enforcement methods, Extensible Authentication Protocol, IEEE 802.X Port based Network Access Control, Access Control Lists.  
 Firewalls: Types of Firewall, Stateless & Stateful Packet Filtering Firewall , Application Gateways , Circuit-Level Gateway, Firewall Deployment with DMZ. Security in internet banking system.

### Texts / References:

| T/R | Book Title/Author                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------|
| T1  | <b>Network Security Essentials: Applications and Standards</b> , 3/e, William Stallings, Pearson, ISBN: 9788131716649. |
| T2  | Network Security Bible, 2 <sup>nd</sup> edition, Eric Cole, Wiley Publisher, ISBN: 978812652331.                       |
| R1  | <b>Network Security and Management</b> , 2 <sup>nd</sup> edition, Brijendra Sing, PHI, ISBN: 9788120339101 .           |

### Online Resources:

| Sl.No | Website Link                                                                                                                                                              |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <a href="https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf">https://www.tutorialspoint.com/network_security/network_security_tutorial.pdf</a> |
| 2     | <a href="https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf">https://training.apnic.net/wp-content/uploads/sites/2/2016/12/TSEC01.pdf</a>           |